

University of Stuttgart
Germany
Institute of Industrial Automation and
Software Engineering

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confidential**

How to train a hamster

Bachelor Thesis 9876

Submitted at the University of Stuttgart by
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Mechatronics

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List of Abbreviations

ISW Institut für Automatisierungstechnik und Softwaresysteme der Universität Stuttgart

SPS Speicherprogrammierbare Steuerung

NDA Non-Disclosure Agreement

Glossary

This section is optional. Ask your supervisor whether it is required for your thesis.

There are two ways to build a glossary:

- If you just want to get it done, then use a `longtable` and fill your names in, see table below.
- If you want it fancy, then package `glossaries` (maybe `glossaries-extra`) may be your way to go. Be warned that although it automates symbol handling (e.g. sorting and referencing of symbols), it comes with some administrative overhead. You can find a discussion on different ways to achieve this on <https://tex.stackexchange.com/a/366282>.
- use the `description-environment` (see below)

Name	Description
Actuator	Component for the conversion of an information carrying, low-energy control signal to a high energy signal capable of process intervention
Name2	Description 2
Name3	Description 3

Actuator Component for the conversion of an information carrying, low-energy control signal to a high energy signal capable of process intervention

Name2 Description 2

Kurzfassung

Deutsche Kurzfassung hier.

Stichwörter: Hamster, Training, Anleitung

Abstract

Add the english abstract here.

Keywords: hamster, training, guide

1. Introduction

The aim of student theses is to show that students are able to work on tasks from their own field independently and according to scientific methods and that they can present their results appropriately. These objectives, defined in the examination regulations, raise a number of questions:

- What does “working according to scientific methods” actually mean?
- What belongs to the proper presentation of results?
- Which standards and guidelines apply at the **IAS!** (IAS!)?
- What is to be considered in general when writing student theses at the **IAS!**?

Read the guidelines for the preparation of student theses at **IAS!**! The chapters used in this template are for inspiration only. Each thesis is different and the chapter headings and their order should be adapted accordingly.

2. Examples

In the following some examples for the conversion in $\text{\LaTeX}$ are listed. For formal requirements, please have a look at the guidelines for the preparation of student theses at the IAS!

2.1. Listings

An introduction for list environments with $\text{\LaTeX}$ can be found at
https://en.wikibooks.org/wiki/LaTeX/List_Structures.

An unordered enumeration can look like this:

- Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.
- Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.
- Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.

Use the `enumerate` environment for ordered lists.

1. Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.
2. Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.

2. Examples

3. Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.

Descriptions are set with the `description` environment.

Mosquito Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.

Emu Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.

Armadillo Fusce tincidunt consectetur nisl a pretium. Nam sed eleifend nunc. Nulla feugiat nisl ac mauris varius, eu viverra tellus condimentum. Nullam tempus dolor a elementum con-vallis. Nam sagittis, nisi non tempor luctus, enim ex pretium nunc, lacinia suscipit arcu augue id sem.

2.2. Cite

Cite with the `\cite{}` command [1], [2]. You can mention authors and titles as well with `\citeauthor{}` and `\citetitle{}`: E.g., `Feuersänger` developed `pgfplots` and describes it in the documentation Manual for Package `pgfplots`.

Use tools for literature management as `JabRef` or `Citavi`.

2.3. Typing math

An introduction for typing math with $\text{\LaTeX}$ can be found at https://de.wikibooks.org/wiki/LaTeX-Kompendium:_F%C3%BCr_Mathematiker. The Wikipedia page for typesetting math is worth a visit as well (<https://de.wikipedia.org/wiki/Formelsatz>).

You can use `\( ... \)` to typeset formulas in text e.g., $\sqrt{x^2}$ or $\begin{bmatrix} a & b & c \end{bmatrix}^T$ (`\bmat` and `\T` are self-defined macros from `settings.tex`).

Multiline math environments can e.g. be set with the `align` environment. A `&` character helps with the vertical alignment.

$$\begin{aligned} c^2 &= a^2 + b^2 \\ \Leftrightarrow c &= \pm \sqrt{a^2 + b^2} \end{aligned} \tag{2.1}$$

2. Examples

Only number equations that will be referenced later, such as Equation 2.3. `\nonumber` suppresses the generation of an equation number.

$$c^2 = a^2 + b^2 \tag{2.2}$$

$$\Leftrightarrow c = \pm \sqrt{a^2 + b^2} \tag{2.3}$$

2.4. Tables

Tables are set in a `table` environment, as shown in Table 2.1. They must always be referenced and explained in the previous text. The package `booktabs` facilitates the consistent use of horizontal line strengths.

Article		
Animal	Description	Price (€)
Mosquito	per gramm	13.65
	per piece	0.01
Gnu	stuffed	92.50
Emu	stuffed	33.33
Armadillo	frozen	8.99

Table 2.1.: Example table using the `booktabs` package

2.5. Graphics

Simple graphics can be integrated with `\includegraphics`. Always use a `figure` environment for graphics. Reference graphics before they appear in the text, such as Figure 2.1, and assign meaningful captions to them.



Figure 2.1.: The logo of the University of Stuttgart

Some graphics with TikZ and PGFplots are shown on Figure 2.2, Figure 2.3a, Figure 2.3b and Figure 2.4 which might be helpful or inspirational for your thesis.

2. Examples

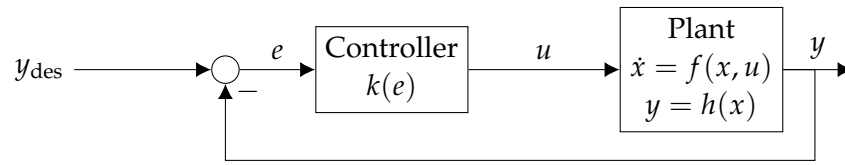
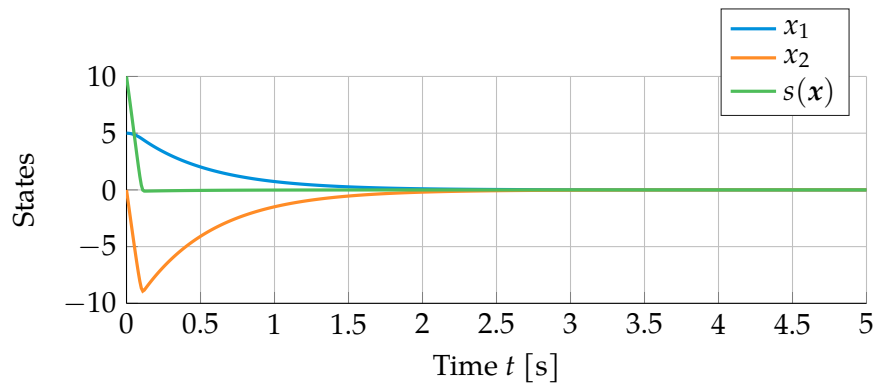
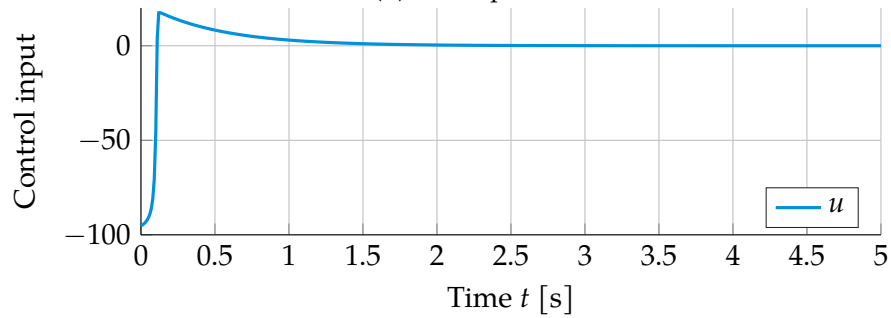


Figure 2.2.: A simple block diagram



(a) States plotted over time



(b) Input $u(\cdot)$ plotted over time

Figure 2.3.: Example for exports from MATLAB with `matlab2tikz`

2. Examples

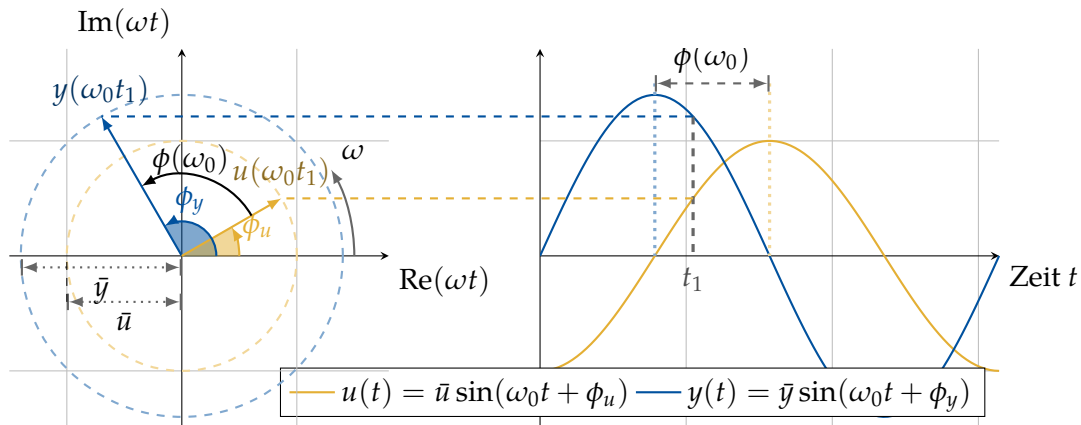


Figure 2.4.: Analytical calculation of a phasor diagram with TikZ [4]

2.6. Code

The listings package, for example, is suitable for code excerpts. Make sure that you only include necessary code, Listing 2.1 is a bad example and should not be included.

```
#include <iostream>

using namespace std;

int main(void){
    cout << "Hello world." << endl;
    return 0;
}
```

Listing 2.1: This code does not provide any insights and should not be included.

2.7. Abbreviations

For abbreviations you can use the package acronym.

The command `\ac{}` introduces the abbreviation at the first use, for example Institut für Automatisierungstechnik und Softwaresysteme (ISW) and several Speicherprogrammierbare Steuerungen (SPS). Then the abbreviations ISW and SPS will be used automatically. The definition of the abbreviations can be done in a separate file, the example document includes chapters/Acronyms.

We recommend resetting the acronym package after the abstract with `\acresetall`, so they will be reintroduced in the introduction chapter.

3. Tooling

3.1. Recommended Editors

There are several options for working on your LaTeX document. All of them have individual strong suits and drawbacks. For locally installed editors it is highly recommended to use version control (see section 3.3).

3.1.1. Overleaf

By far the easiest and fastest setup is provided by `overleaf.com`. After creating an account, you can add this template and start working on your thesis right away. Overleaf allows simultaneous editing of by multiple persons and provides integrated version control and document compilation. However, sometimes it is not possible to have your work hosted on public servers due to Non-Disclosure Agreements (NDAs).

Ask your supervisor if you're allowed to use overleaf!

3.1.2. Visual Studio Code

VSCode is also a valid choice for editing latex documents. Just install the LaTeX Workshop extension. You should also consider installing the LTeX extension for integrating LanguageTool (see section 3.2).

3.1.3. TexStudio

You can use the TeXstudio¹ latex editor and the MiKTeX² LaTeX distribution on your PC.

In TeXstudio, remember to set `lualatex` as the standard compiler and `biber` as the `bibtex` backend.

3.1.4. direct command

If you use the `latex` or `pdflatex` commands directly please remember that you need to run the commands in this order:

1. `pdflatex Name_Thema.tex`

¹<https://www.texstudio.org/>

²<https://miktex.org/>

3. Tooling

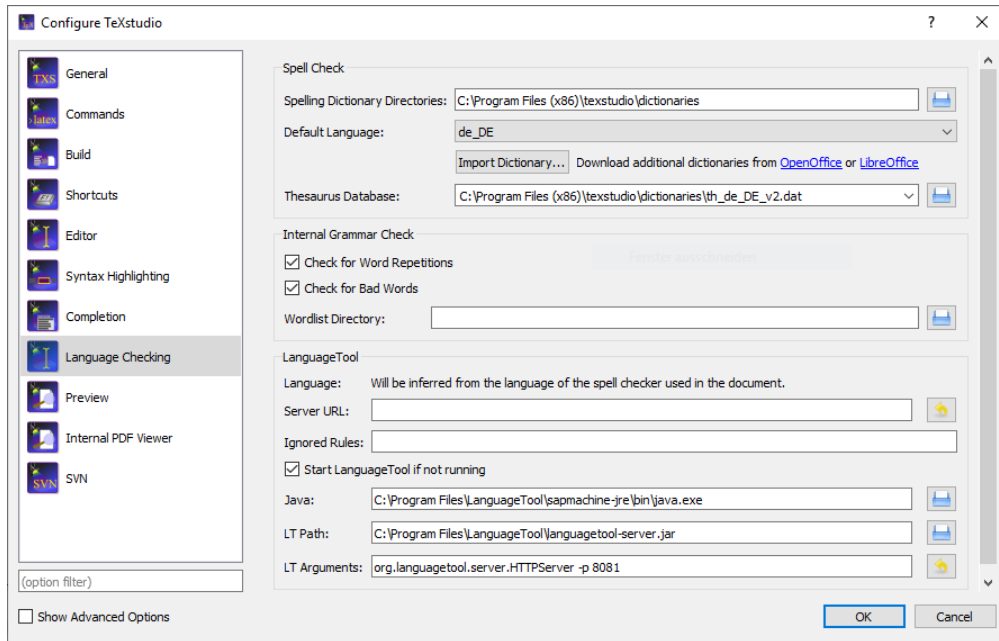


Figure 3.1.

2. `biber Name_Thema`
3. `pdflatex Name_Thema.tex`
4. `pdflatex Name_Thema.tex`

This is necessary that latex knows all your references and can insert them correctly. For more details see: <https://tex.stackexchange.com/questions/53235/why-does-latex-bibtex-need-three-passes-to-clear-up-all-warnings#>.

3.2. Checking of grammar, spell and style

As with all word-processing tasks, it is highly recommended to use at least some sort of spell-checking software. For this purpose TexStudio provides integration with LanguageTool³. LanguageTool is a java based application, so make sure to install a Java Runtime Environment. A exsample setting for your PC are shown in Figure 3.1. Further information on the integration of LanguageTool with TexStudio can be found at the LanguageTool Wiki⁴.

³<https://languagetool.org>

⁴<http://wiki.languagetool.org/checking-la-tex-with-languagetool#toc2>

3.3. Version control

Whenever working on a document, it is desirable to have some sort of version control.

For this task, you can use for example github <https://github.com>. Once you are logged in to github, you can create your own repository for tracking your thesis files. A .gitignore file is necessary to not track all changes to automatically generated files.

You may use the .gitignore provided by github

<https://github.com/github/gitignore/blob/main/TeX.gitignore>.

You can achieve a rudimentary integration with TeXstudio by defining macros such as the git commit macro shown in Listing 3.1. Note that macros can also be called by using shortcuts.

```
%SCRIPT
dialog = new UniversalInputDialog()
dialog.setWindowTitle("Git commit")
dialog.add("", "Message", "comment")
dialog.add(false, "Commit all files", "allfiles")
if (dialog.exec() != null) {
comment = dialog.get("comment")
if ((dialog.get("allfiles")) == true){
buildManager.runCommand(
"git commit -a -m \"" + comment + "\"", editor.fileName())
}else{
buildManager.runCommand(
"git commit " + editor.fileName() + " -m \"" + comment +
"\"", editor.fileName())
}
}
```

Listing 3.1: git commit macro

4. Conclusions

Bibliography

- [1] T. Cubitt, *The cleveref package*, 2013.
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- [13] N. Talbot, *User manual for glossaries.sty v4.09*, 2014.

A. Example appendix chapter

Declaration of Compliance

I hereby declare to have written this work independently and to have respected in its preparation the relevant provisions, in particular those corresponding to the copyright protection of external materials. Whenever external materials (such as images, drawings, text passages) are used in this work, I declare that these materials are referenced accordingly (e.g. quote, source) and, whenever necessary, consent from the author to use such materials in my work has been obtained.

Signature:

Stuttgart, on the 10 March 2020